

ROLLESTON, Australia

WMO: 943960

Lat: 24.4615S

Lon: 148.6260E

Elev: 220

StdP: 98.71

Time Zone: 10.00 (E10)

Period: 10-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	3.2	4.6	-7.8	2.0	18.3	-5.6	2.4	16.4	7.0	16.2	6.3	17.2	1.3	150	0.350

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.7	38.0	20.9	36.5	20.7	35.1	20.6	24.5	30.0	23.8	29.3	23.3	28.9	3.1	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.3	18.5	26.3	22.5	17.7	25.9	21.8	16.9	25.4	75.5	29.4	72.7	29.2	70.6	28.9	26.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.0	6.2	5.5	DB	0.5	40.4	0.9	2.1	-0.1	41.9	-0.6	43.1	-1.1	44.3	-1.7	45.8	(4)
(5)			WB	-1.2	25.4	1.4	0.8	-2.2	26.0	-3.0	26.4	-3.8	26.9	-4.8	27.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	21.7	27.8	26.9	25.5	21.9	18.1	15.2	14.8	16.0	19.5	22.7	25.5	27.1	(6)	
	DBStd	5.36	2.23	2.51	2.23	2.15	2.78	2.66	2.78	2.66	3.04	2.88	2.68	2.54	(7)	
	HDD10.0	1	0	0	0	0	0	0	1	0	0	0	0	0	(8)	
	HDD18.3	356	0	0	0	3	39	97	113	82	20	3	0	0	(9)	
	CDD10.0	4283	553	473	481	358	249	157	151	187	286	393	466	531	(10)	
	CDD18.3	1597	294	240	222	111	30	4	4	11	55	138	216	272	(11)	
	CDH23.3	17727	3474	2732	2176	994	308	40	46	190	750	1504	2376	3137	(12)	
(13)	CDH26.7	8239	1829	1371	951	286	44	2	2	24	258	623	1182	1667	(13)	
(14)	Wind	WSAvg	2.5	2.9	2.9	2.7	2.4	2.2	2.1	2.0	2.2	2.3	2.5	2.6	2.7	(14)
(15) Precipitation	PrecAvg	621	94	83	57	37	41	22	25	25	24	45	63	100	(15)	
	PrecMax	991	325	275	360	265	179	105	160	157	102	180	165	341	(16)	
	PrecMin	249	12	0	0	0	0	0	0	0	0	0	0	17	(17)	
	PrecStd	175	64	63	64	49	40	26	30	34	29	40	37	70	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.0	39.9	37.0	32.6	29.7	25.9	26.3	28.9	35.7	35.9	37.8	39.6	(19)	
		MCWB	22.0	20.9	21.2	19.6	18.2	16.4	16.5	15.6	17.5	18.5	20.2	20.0	(20)	
	2%	DB	37.4	37.5	34.3	30.8	27.9	24.5	24.5	27.1	31.9	33.4	35.8	37.5	(21)	
		MCWB	21.6	20.9	21.0	19.1	17.1	15.9	15.2	15.7	16.4	18.7	19.7	20.4	(22)	
	5%	DB	35.9	35.2	33.0	29.5	26.5	23.2	23.2	25.5	29.5	31.8	34.2	35.5	(23)	
		MCWB	21.6	21.4	20.9	18.9	16.7	15.1	14.7	14.7	15.8	18.2	19.4	20.4	(24)	
	10%	DB	34.2	33.2	31.6	28.4	25.0	21.9	21.8	23.9	27.4	30.2	32.7	33.9	(25)	
		MCWB	21.5	21.3	20.9	18.3	16.2	14.2	13.9	13.5	15.6	17.7	19.4	20.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.4	25.2	24.6	21.7	20.2	18.1	19.3	19.4	20.6	22.0	23.2	24.5	(27)	
		MCDB	32.0	30.3	28.6	28.4	26.9	22.6	23.9	25.0	24.0	29.1	30.7	30.8	(28)	
	2%	WB	24.6	24.4	23.7	21.0	18.8	17.1	17.9	17.6	19.5	21.1	22.2	23.4	(29)	
		MCDB	30.8	29.5	27.8	26.9	24.2	22.6	21.6	22.9	23.6	28.0	30.7	28.9	(30)	
	5%	WB	23.9	23.6	23.1	20.2	17.8	16.2	16.7	16.4	18.5	20.4	21.4	22.7	(31)	
		MCDB	29.9	29.4	28.3	26.2	24.0	21.2	20.5	22.2	24.3	27.4	29.5	28.5	(32)	
	10%	WB	23.3	23.0	22.5	19.6	17.1	15.4	15.4	15.1	17.6	19.7	20.7	22.1	(33)	
		MCDB	29.1	29.2	27.8	25.6	23.3	20.0	19.5	21.4	24.6	26.5	28.2	28.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	11.7	11.7	11.0	13.2	14.1	13.3	14.7	16.7	15.7	14.5	13.0	12.1	(35)	
		MCDBR	14.3	14.6	13.5	14.5	15.2	14.7	15.9	18.3	18.4	16.6	15.4	14.7	(36)	
	5% WB	MCWBR	3.5	3.8	3.6	5.2	6.6	7.0	8.0	8.8	6.9	5.5	4.6	3.9	(37)	
		MCWBR	10.8	10.4	9.4	11.9	12.6	11.5	11.0	14.5	12.7	12.6	12.5	10.4	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.380	0.364	0.346	0.329	0.309	0.300	0.291	0.306	0.340	0.369	0.389	0.387	(40)	
		taud	2.517	2.575	2.614	2.620	2.614	2.650	2.655	2.585	2.490	2.441	2.430	2.474	(41)	
	Ebn at Noon		963	965	956	929	909	899	921	940	945	948	949	958	(42)	
		Edh at Noon	114	105	96	88	81	75	76	89	107	119	123	119	(43)	
(44) All-Sky Solar Radiation	RadAvg		6.85	6.49	5.83	5.13	4.25	3.64	4.08	5.02	5.99	6.69	6.98	7.04	(44)	
	RadStd		0.52	0.56	0.47	0.43	0.22	0.26	0.31	0.31	0.52	0.51	0.77	0.71	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page